

Kaiyu He

917-238-2465 | kaiyu.he@utdallas.edu | GitHub | Homepage | Google Scholar
Research interests: Large Language Models, Language Agents, Reasoning & Learning, Evaluation

SUMMARY

Ph.D. student in Computer Science (GPA 4.0/4.0) specializing in **LLM reasoning, agentic evaluation, statics and machine learning**. I lead end-to-end research—from problem framing and benchmark design to large-scale experiments, RL post-training, and open-source releases. Multiple first-/co-author works appear in the **ACL/EMNLP/TMLR/ICLR** ecosystem. Built a **general abductive reasoning evaluation framework** and an **interactive rule-learning environment**, delivering consistent gains across strong open-source models.

EDUCATION

University of Texas at Dallas

Ph.D. in Computer Science; GPA: 4.0/4.0

Aug 2023–Jun 2027 (expected)

Dallas, TX

Columbia University

M.S. in Biostatistics; GPA: 4.0/4.4

Sep 2021–Jun 2023

New York, NY

Renmin University of China

B.S. in Applied Statistics; GPA: 3.2/4.0

Sep 2016–Jun 2020

Beijing, China

RESEARCH EXPERIENCE

Human Language Technology Research Institute, UTD

Research Assistant (Advisor: [Zhiyu Zoey Chen](#))

May 2023–Present

Dallas, TX

- Core focus on **explainability, reasoning, and learning** for LLM agents across abductive/deductive/inductive paradigms; designed evaluation frameworks and training algorithms to shift from “solving” to “**learning to solve**.”
- **GEAR**: First general evaluation & training framework for abductive reasoning. Ran large-scale evaluation over **9 LLMs** and **4 benchmarks**; RL/process-reward training boosted downstream Top-3 accuracy on open-source models (**Llama-3.1-8B: 13×**, **NextCoder-7B: 1.7×**, **Qwen-2.5-7B: 1.4×**). *Released code/data for reproducibility.*
- **IDEA & RULEARN**: Built an **extensible, interactive** text environment and agent framework for rule learning; achieved **+10%** puzzle success and **-30.2%** repetitive actions vs. ReAct across **5 LLMs**.
- Led research ops: multi-GPU/distributed pipelines, principled metrics (consistency, generalizability, diversity), rigorous visualization, and **publication-quality** figures/tables and open-source hygiene.

Columbia NLP Group, Columbia University

Research Assistant (Advisor: [Kathleen McKeown](#))

Jan 2023–Jun 2023

New York, NY

- Designed multimodal models for complex event detection in **long, untrimmed video**; delivered robust gains in temporal modeling and retrieval relevant to agentic RAG pipelines.

Mailman School of Public Health, Columbia University

Research Assistant (Advisor: [Sen Pei](#))

Apr 2022–Jun 2023

New York, NY

- Analyzed **740k+** NYC contact-tracing calls; proposed counterfactual intervention that can raise completion rates by up to **7.8%**. Result published in **BMC Public Health**.

Vagelos College of Physicians & Surgeons, Columbia University

Research Assistant (Advisor: [Xiaofu He](#))

Jan 2022–Aug 2022

New York, NY

- Built a real-time **Muse2 EEG** decoded neurofeedback platform with near-real-time signal processing and visualization; presented at **rtFIN 2022** (talk + poster).

PUBLICATIONS & PREPRINTS (FIRST AUTHOR)

GEAR: A General Evaluation Framework for Abductive Reasoning

First automatic evaluation & training framework for LLM abductive reasoning

In submission to ICLR 2026

Long Paper

- Roles: Idea, coding, experiment design, RL algorithm. [[Paper](#) | [Code](#)]

• Evaluated **9 LLMs/4 benchmarks**; consistent Top-3 gains on open-source models (**Llama-3.1-8B 13×**, **NextCoder-7B 1.7×**, **Qwen-2.5-7B 1.4×**).

From Reasoning to Learning: A Survey on Hypothesis Discovery and Rule Learning with LLMs

TMLR

*Awarded the **TMLR Survey Certification***

Survey

- Roles: Literature review, framework design, writing. [[Paper](#) | [Reviews](#)]
- First comprehensive framework grounded in Peirce's abduction–deduction–induction; clarifies gaps and future directions for hypothesis discovery.

IDEA: Enhancing the Rule Learning Ability of Large Language Model Agent

ACL 2025 Findings

Interactive rule-learning environment and agent framework

Long Paper

- Roles: Benchmark design (**RULEARN**), framework (**IDEA**), coding, experiments. [[Paper](#) | [Code](#)]
- **+10%** puzzle success; **-30.2%** repetitive actions vs. ReAct across **5 LLMs**.

Evaluating completion rates of COVID-19 contact tracing surveys in NYC

BMC Public Health

Demographic drivers; counterfactual policy improves completion

Long Paper

- Roles: Data analysis, statistical modeling, writing. [[Paper](#)]
- Dataset: **740k+** calls; proposed policy increases completion by up to **7.8%**.

SELECTED COLLABORATION PUBLICATIONS & POSTERS

HiPRAG: Hierarchical Process Rewards for Efficient Agentic RAG

In submission to ICLR 2026

Hierarchical reward shaping for efficient Agentic RAG

Long Paper

- Roles: Manuscript drafting, data analysis.

LMR-BENCH: Evaluating LLM Agents on Reproducing NLP Research

EMNLP 2025

A benchmark for agentic reproducibility in NLP research

Long Paper

- Roles: Drafting, benchmark construction, analysis, visualization. [[Paper](#)]

A Decoded EEG Neurofeedback Platform Using Muse2

rtFIN 2022

Real-time EEG neurofeedback platform

Poster & Talk

- Roles: Model design, preprocessing, coding, training. [[Poster](#)]

PROJECTS (SELECTED)

Latent Space Discovery of DCGAN

May 2022–Jun 2022

Personal Project

GAN, Representation

- Trained DCGAN and visualized **3D latent structure** to interpret semantic traversals. [[Code](#)]

Abstractive Summarization with SimCLS

Oct 2022–Dec 2022

Personal Project

NLP Summarization

- Adapted SimCLS for medical abstracts/title generation; validated across multiple models. [[Blog](#) | [Code](#)]

VR Educational Game (Columbia Innovation Award)

Mar 2022–May 2022

Teachers College, Columbia University

Unity, VR

- Co-built a physics education VR game improving interactive learning (demo [video](#); [code](#)).

ADDITIONAL EXPERIENCE & SKILLS

Game Numeric Designer, 89 Trillion (Art of War: Legions)

Aug 2020–Apr 2021

Industry Experience

Beijing, China

- Partnered with the AI team on **numerics & recommendation** features; led A/B testing and content balancing; shipped stable versions to **millions of users** (sample [video](#)).

Tooling & Platforms: PyTorch, TensorFlow, scikit-learn, Transformers, Unity

Programming: Python, C#, C/C++, R, SQL, Shell, $\text{\LaTeX}$

Research Ops: multi-GPU/distributed evaluation, RL post-training (e.g., DPO, process rewards), visualization, reproducible open-source practice

Certification: TensorFlow Developer Certificate